Type MS · Explain how airway features keep air clean and flowing

Answer 1 of 5 · Question 6 part b · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>Increase/goes up;</td><td>allow equivalent</td><td>1</td></tr>
<tr><td>(ii)</td><td>Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;;</td><td>Allow one mark for reference to lifestyles only</td><td>2</td></tr>
<tr><td>(b)</td><td>Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed;
Increased risk of (lung/respiratory) infection/coughing;</td><td></td><td>1
1</td></tr>
<tr><td>(c) (i)</td><td><table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table></td><td>Allow for SA:V ratio 6:3 ECF</td><td>3</td></tr>
<tr><td>(ii)</td><td>Smaller surface area to volume ratio (in lungs with emphysema);
Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide;
Less oxygenated blood/less oxygen transferred (to cells);
Less (aerobic) respiration;</td><td></td><td>Max 3</td></tr>
</table>

Type MS · Explain how airway features keep air clean and flowing

Answer 1 of 5 · Question 6 part b · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how airway features keep air clean and flowing

Answer 2 of 5 · Question 6 part b.iii · 2 marks

2020 JAN · Paper 01 · find part b.iii in this mark scheme extract

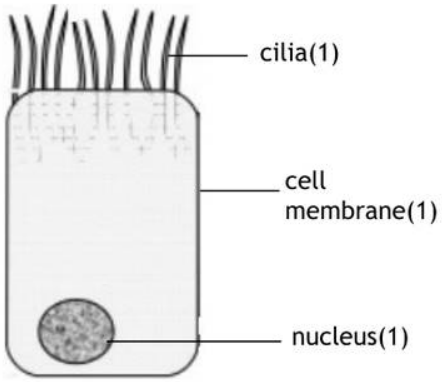
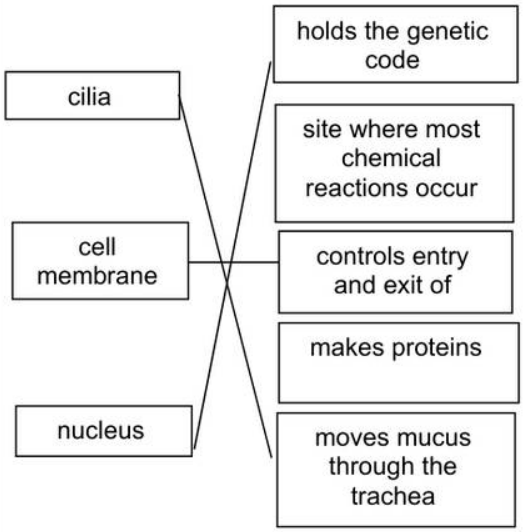
Question number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · Explain how airway features keep air clean and flowing

Answer 3 of 5 · Question 2 part b.ii · 2 marks

2020 NOV · Paper 01 (regional R) · find part b.ii in this mark scheme extract

2 (a) (i)			3
(ii)		Reject more than one line from each structure	3
(b) (i)	Tar (1)		1
(ii)	• (cilia)burned/destroyed/reduced in number(1) • paralysed/cannot beat to and fro/move mucus(1)		2

Total for Question 2 = 9 Marks

Type MS · Explain how airway features keep air clean and flowing

Answer 4 of 5 · Question 2 part a.i · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
2	(a) (i)	an answer that makes reference to three of the following: • warms air; • moistens air; • layer of (sticky) mucus; • removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs); • reference to 1st line of defence;	Mp3 must be linked to mp4 for mark	3
	(ii)	• prevents collapse; • to allow air to flow freely (to lungs/alveoli); • allows neck to bend;		2
	(iii)	B; (bronchi)		1
	(b)	D; (medulla oblongata)		1
	(c)	$17 \times 0.50 = 8.5\ dm^3$; 4×8.5; $= 0.34\ dm^3$;		3
	(d)	• reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin; • less oxygen to developing fetus; • lower rate of respiration/less energy released (by fetus); • less growth/lower birth mass;		4

Total 14 marks

Type MS · Explain how airway features keep air clean and flowing

Answer 4 of 5 · Question 2 part a.i · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how airway features keep air clean and flowing

Answer 5 of 5 · Question 2 part a.ii · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
2	(a) (i)	an answer that makes reference to three of the following: • warms air; • moistens air; • layer of (sticky) mucus; • removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs); • reference to 1st line of defence;	Mp3 must be linked to mp4 for mark	3
	(ii)	• prevents collapse; • to allow air to flow freely (to lungs/alveoli); • allows neck to bend;		2
	(iii)	B; (bronchi)		1
	(b)	D; (medulla oblongata)		1
	(c)	$17 \times 0.50 = 8.5\ dm^3$; 4×8.5; $= 0.34\ dm^3$;		3
	(d)	• reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin; • less oxygen to developing fetus; • lower rate of respiration/less energy released (by fetus); • less growth/lower birth mass;		4

Total 14 marks

Type MS · Explain how airway features keep air clean and flowing

Answer 5 of 5 · Question 2 part a.ii · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks